

Assignment 2

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Question 1

Table 1: Training accuracies with different hyper-parameters

λ	ExpLinear			Hinge		
	1	2	3	1	2	3
0.001	0.9994	0.594	0.8772	0.9982	0.6116	0.8728
0.01	0.9988	0.5987	0.8761	0.9958	0.6135	0.8704
0.1	0.9957	0.6071	0.8755	0.9926	0.6195	0.87
1.0	0.9904	0.5981	0.8685	0.9826	0.6169	0.8634

Table 2: Test accuracies with different hyper-parameters

λ	ExpLinear			Hinge		
	1	2	3	1	2	3
0.001	0.98987	0.57165	0.88179	0.98853	0.6027	0.8728
0.01	0.98956	0.56721	0.88049	0.98712	0.60148	0.87741
0.1	0.98642	0.57839	0.88125	0.98358	0.59695	0.87768
1.0	0.98027	0.57372	0.87373	0.97336	0.60421	0.87106

e) From the results, both ExpLinear and Hinge classifiers performed very well on Model 1, showing near-perfect accuracy since the data is likely linearly separable. For Model 2, accuracies were much lower across all λ values, suggesting the data is non-linear and can't be well captured by a linear boundary. Model 3 showed intermediate performance with stable results. As λ increased, both training and test accuracies slightly decreased, which makes sense because stronger regularization reduces overfitting but can slightly limit the model's flexibility. Overall, ExpLinear achieved slightly higher training accuracy, but Hinge was a bit more stable and generalized better on test data. The small differences between the two show that both are effective, but Hinge tends to be more robust, especially when data is noisy or not perfectly separable.

Question 2

Table 3: Training accuracies with different kernel

Kernel	ExpLinear			Hinge		
	1	2	3	1	2	3
Linear	0.999	0.58	0.877	0.999	0.602	0.873
Poly(d=2)	1	1	0.875	1	1	0.876
Poly(d=3)	1	1	1	1	1	1
Gauss($\sigma = 1$)	1	1	1	1	1	1
Gauss($\sigma = 0.5$)	1	1	1	1	1	1

Table 4: Test accuracies with different kernels

Kernel	ExpLinear			Hinge		
	1	2	3	1	2	3
Linear	0.988	0.5733	0.879	0.986	0.5999	0.8761
Poly(d=2)	0.9849	0.9791	0.8836	0.9832	0.9709	0.8815
Poly(d=3)	0.9833	0.9631	0.995	0.9813	0.9613	0.992
Gauss($\sigma = 1$)	0.9852	0.9775	0.9967	0.9787	0.9682	0.9961
Gauss($\sigma = 0.5$)	0.9637	0.9589	0.9986	0.9986	0.9509	0.998

e) As we can see Model 1 achieves near perfect performance in both train and test. This is because this model is simple, as we can see the linear kernel is sufficient for this model and for more complex models the accuracy drops. At more complex seconds model, we can see tht Linear kernel fails. This is because model 2 has non convex boundary while non linear kernels work fine with it. We can also see that Gauss($\sigma = 0.5$) has lowest accuracy on test in Model 1 and 2. Small σ allows boundary to capture noise in small training set.

Question 3

c)

Cross-validation accuracy matrix

0.79487179

0.58974359

0.58974359

0.58974359

0.79487179

0.58974359

Best λ : 0.1

Best Kernel : linearKernel